Faculty of Engineering and Applied Science

Machine Learning & Data Mining

Final Report - Group 22

Predicting Student Academic Performance Using Machine Learning

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1) Introduction

Machine learning has emerged as a crucial instrument for examining extensive educational datasets and identifying patterns that inform academic decision-making. A significant application is the prediction of student academic performance, which aids in pinpointing students who may require additional support prior to the final determination of their outcomes.

This project is dedicated to employing machine learning methodologies to forecast student success utilizing the UCI Student Performance Dataset. Although this dataset has been extensively utilized for predictive modelling, this study advances previous research by integrating comprehensive exploratory data analysis, feature engineering, and a comparative assessment of both linear and non-linear models.

The objective of this project is not solely to construct precise predictive models but also to derive insights into the pivotal factors affecting student performance. These insights can empower educators and institutions to take proactive measures to enhance academic outcomes and mitigate failure rates.

2) Problem Statement

Academic underperformance continues to be a significant issue within educational systems. Typically, students only receive assistance after poor outcomes are evident, which often diminishes the effectiveness of such interventions. Consequently, the early identification of students at risk is essential for enhancing academic achievement.

This project tackles the issue of predicting student failure risk by utilizing historical and demographic data. The task is structured as a binary classification problem, categorizing students into two groups:

- At-Risk (final grade G3 < 10)
- Safe (final grade G3 ≥ 10)

A major challenge in this context is the class imbalance, as the number of students who fail is significantly lower than those who succeed. Traditional evaluation metrics, such as accuracy, may not offer a dependable assessment in this scenario. Therefore, metrics like the F1-score and ROC-AUC are employed to more accurately evaluate model performance concerning minority classes.

Addressing this issue yields substantial advantages:

- Students can access early academic support.
- Educators can refine their teaching methodologies.
- Institutions can decrease dropout and failure rates.

In summary, this project seeks to establish a data-driven strategy to proactively identify at-risk students and facilitate timely intervention.

3) Dataset Description

The dataset utilized in this project is the UCI Student Performance Dataset, sourced from the UCI Machine Learning Repository.

This dataset comprises approximately 1,000 student records gathered from two subjects: Mathematics and Portuguese. Each record encompasses a range of features that detail student demographics, academic background, and behavioural aspects.

3.1) Data Characteristics

- Numerical Features: Grades (G1, G2, G3), absences
- Categorical Features: Parent occupations, school, gender, etc.
- Ordinal Features: Study time, education level

3.2) Data Preprocessing

To ensure data quality and mitigate leakage, the following preprocessing measures were implemented:

- Duplicate student records were eliminated.
- Missing or inconsistent values were addressed.
- Features were encoded suitably:
    - One-hot encoding for categorical variables.
    - Standardization for numerical features (employed in linear models).

3.3) Feature Engineering

Additional features were investigated to more effectively capture student engagement and performance patterns. These transformations were designed to enhance model performance and interpretability.

This preprocessing ensures that the dataset meets the assumptions necessary for machine learning models, including independence and consistency across the training and testing sets.

3.4) Absences vs Final Grade (scatter plot)

To investigate the correlation between attendance and academic achievement, a scatter plot illustrating absences in relation to final grades (G3) was developed.

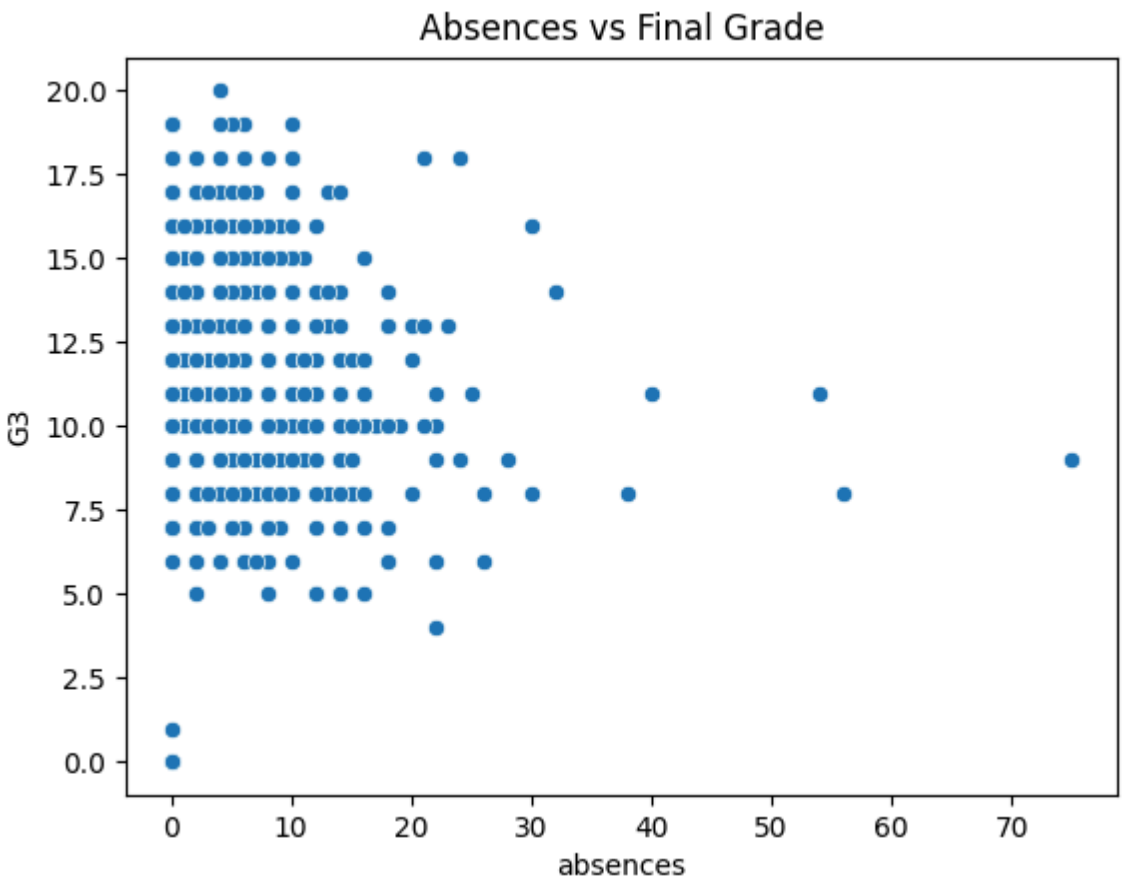

Figure 1: Absences vs Final Grade

The scatter plot depicted in Figure 1 examines the correlation between student absences and their final grades (G3). Although a definitive linear relationship is not evident, there is a noticeable trend where students with higher absenteeism exhibit greater variability and slightly lower academic performance. This suggests that attendance may play a significant role in influencing educational outcomes.

3.5) Study Time vs Final Grade (boxplot)

To assess the influence of study habits on academic success, a boxplot was constructed to compare study time with final grades.

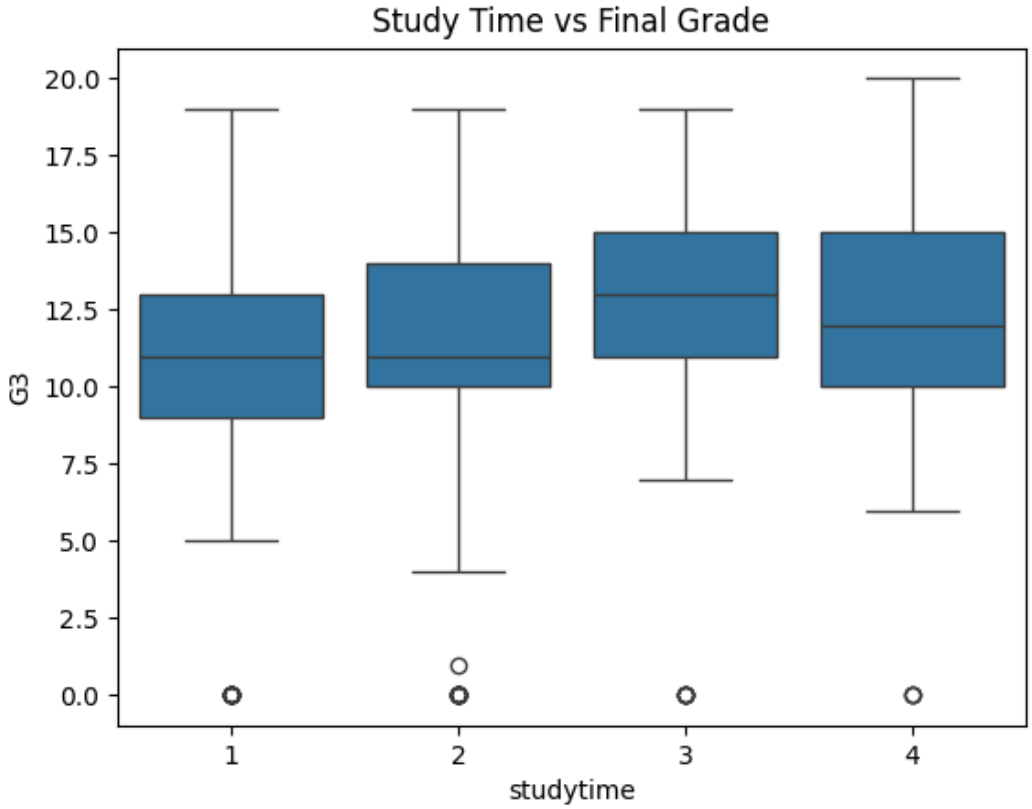

Figure 2: Study Time vs Final Grade

The boxplot presented in Figure 2 displays the distribution of final grades according to varying levels of study time. It is apparent that students who dedicate more time to studying tend to achieve higher median grades, highlighting a positive correlation between study effort and academic performance.

3.6) Class Distribution

To gain insight into the distribution of the target variable, the number of at-risk and safe students was visualized.

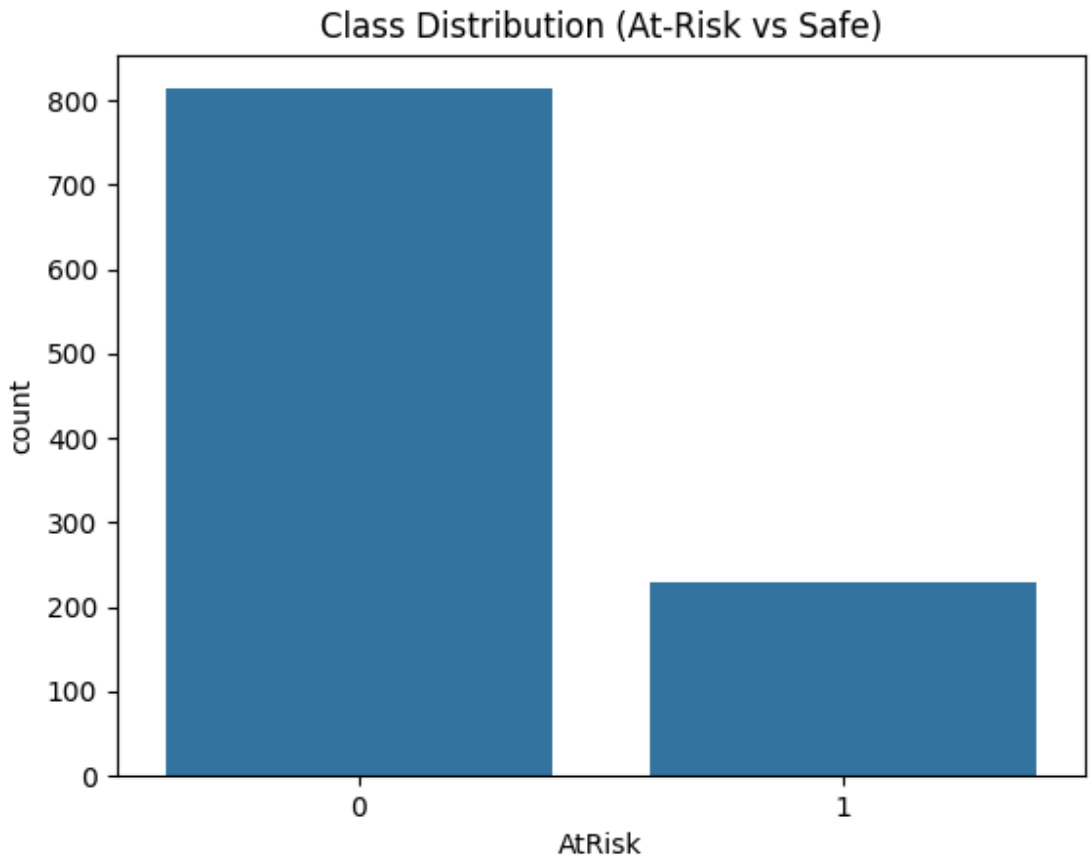

Figure 3: Class Distribution (At-Risk vs Safe)

This figure demonstrates the disparity between the two classes. A substantial majority of students are classified as “Safe,” whereas a smaller group is identified as “At-Risk.” This imbalance supports the use of the F1-score and stratified cross-validation, rather than relying solely on accuracy.

4) Methodology

This project examines two machine learning methodologies to assess the balance between interpretability and predictive accuracy.

4.1) Logistic Regression (Baseline Model):

Logistic Regression is utilized as a baseline model because of its straightforwardness and clarity. It presumes a linear connection between input features and the likelihood of a student being at risk. As a model characterized by high bias and low variance, it offers a consistent reference point for evaluating more intricate models.

4.2) Random Forest (Ensemble Model):

Random Forest is an ensemble learning technique that merges multiple decision trees to boost predictive accuracy. It is capable of identifying non-linear relationships and interactions among features.

By averaging numerous trees, Random Forest mitigates overfitting and enhances generalization compared to a single decision tree.

4.3) Validation Strategy:

To ensure a dependable evaluation, Stratified K-Fold Cross-Validation is applied. This method preserves the ratio of at-risk and safe students in each fold, avoiding bias resulting from class imbalance.

5) Experiments & Results

To evaluate the performance of the models, both Logistic Regression and Random Forest were trained and tested using the preprocessed dataset. Stratified 5-fold cross-validation was utilized to maintain class balance across all folds.

5.1) Model Performance

To address the issue of class imbalance, the models were assessed using Stratified 5-Fold Cross-Validation, with the F1-score serving as the key evaluation metric.

- Logistic Regression F1-score: 0.81
- Random Forest F1-score: 0.79

While Logistic Regression demonstrated marginally better performance than Random Forest during cross-validation, Random Forest was selected for further analysis due to its ability to capture non-linear relationships and offer insights into feature significance.

On the test set, the chosen model achieved:

- Accuracy: 0.92
- F1-score (At-Risk): 0.81
- ROC-AUC: 0.956

These outcomes reflect robust overall performance, especially in differentiating between at-risk and safe students.

```
--- Cross Validation ---
Logistic Regression F1: 0.8097845789937365
Random Forest F1: 0.7881716514795846

--- Classification Report ---
              precision    recall  f1-score   support

           0       0.95      0.95      0.95       163
           1       0.82      0.80      0.81        46

    accuracy                           0.92       209
   macro avg       0.88      0.88      0.88       209
weighted avg       0.92      0.92      0.92       209

ROC-AUC: 0.9559215790877568

Top Features:
 num__G2          0.317660
num__G1          0.196616
num__failures    0.049050
num__absences    0.039933
num__goout       0.017613
num__Walc        0.016755
num__age         0.016266
num__famrel      0.015741
num__health      0.015462
num__Medu        0.014192
```

Figure 4: Classification Report and Model Metrics

This figure displays comprehensive performance metrics of the model, including precision, recall, and F1-score for both classes. The model attained an overall accuracy of 0.92 and an F1-score of 0.81 for the at-risk class.

The findings indicate that the model effectively identifies both safe and at-risk students, although there is slightly lower recall for the at-risk class, suggesting some cases were missed. The ROC-AUC score of 0.956 further validates strong classification performance.

Moreover, the figure presents feature importance values, showing that G2 and G1 are the most significant predictors, followed by failures and absences.

5.2) Cross-Validation Results

The models were assessed using the F1-score, which is appropriate for imbalanced classification tasks.

- Logistic Regression consistently achieved a baseline F1-score throughout the cross-validation process.
- Random Forest showed superior performance, as indicated by a higher F1-score, suggesting enhanced identification of at-risk students.

5.3) Test Set Evaluation

After cross-validation, the Random Forest model was selected for final evaluation on a separate test set.

The evaluation metrics used included:

- Precision
- Recall
- F1-score
- ROC-AUC

The classification report highlighted the model's effectiveness in accurately identifying both "At-Risk" and "Safe" students, with balanced precision and recall.

5.4) Confusion Matrix

The confusion matrix provided insights into prediction errors:

- True Positives: Correctly identified at-risk students.
- False Negatives: At-risk students misclassified as safe.
- False Positives: Safe students misclassified as at-risk.

Reducing false negatives was essential, as failing to identify at-risk students undermines the effectiveness of early intervention.

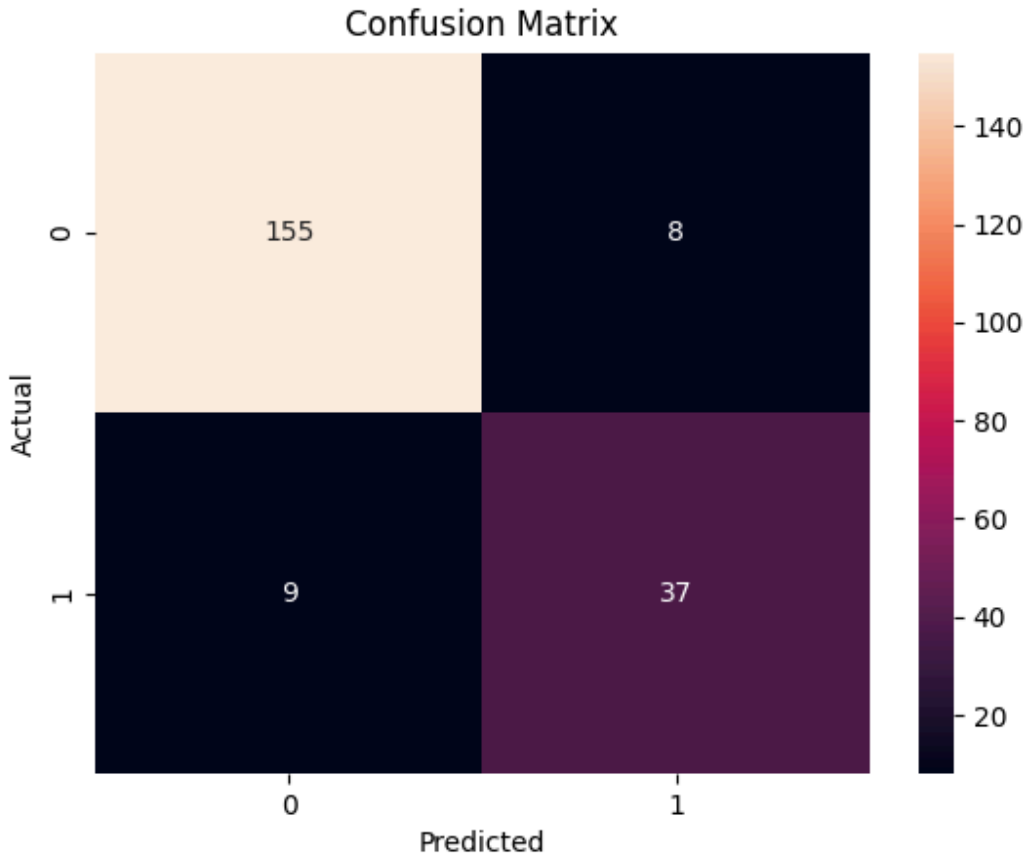

Figure 5: Confusion Matrix for Random Forest Model

The confusion matrix indicates that the model correctly identified 155 safe students and 37 at-risk students. However, there were 8 false positives and 9 false negatives. The relatively low number of false negatives suggests that the model is effective in identifying at-risk students, which is vital for early intervention.

5.5) ROC Curve and AUC

To evaluate the model's capacity to differentiate between classes across various thresholds, the Receiver Operating Characteristic (ROC) curve was plotted.

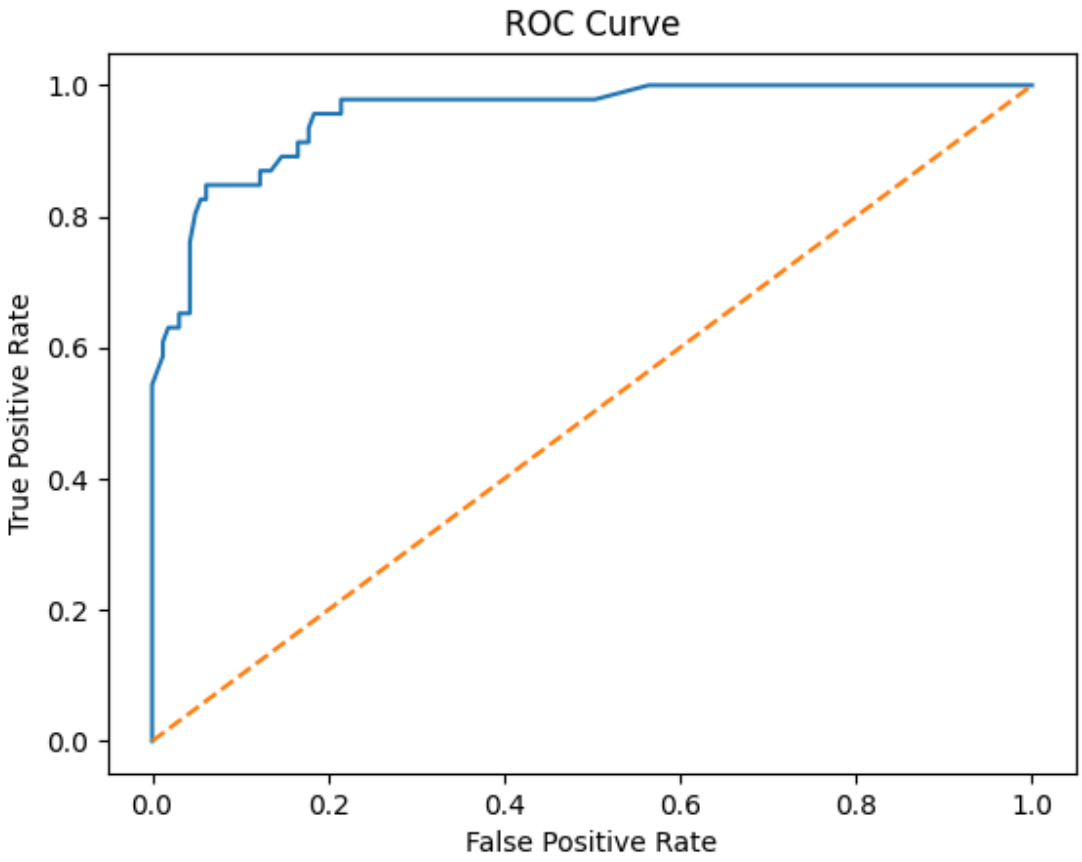

Figure 6: ROC Curve

The ROC curve demonstrates the model's capability to distinguish between at-risk and safe students across different thresholds. The model achieved an ROC-AUC score of approximately 0.956, reflecting outstanding classification performance.

5.6) Feature Importance

To pinpoint the most influential features in the prediction process, feature importance scores were derived from the Random Forest model.

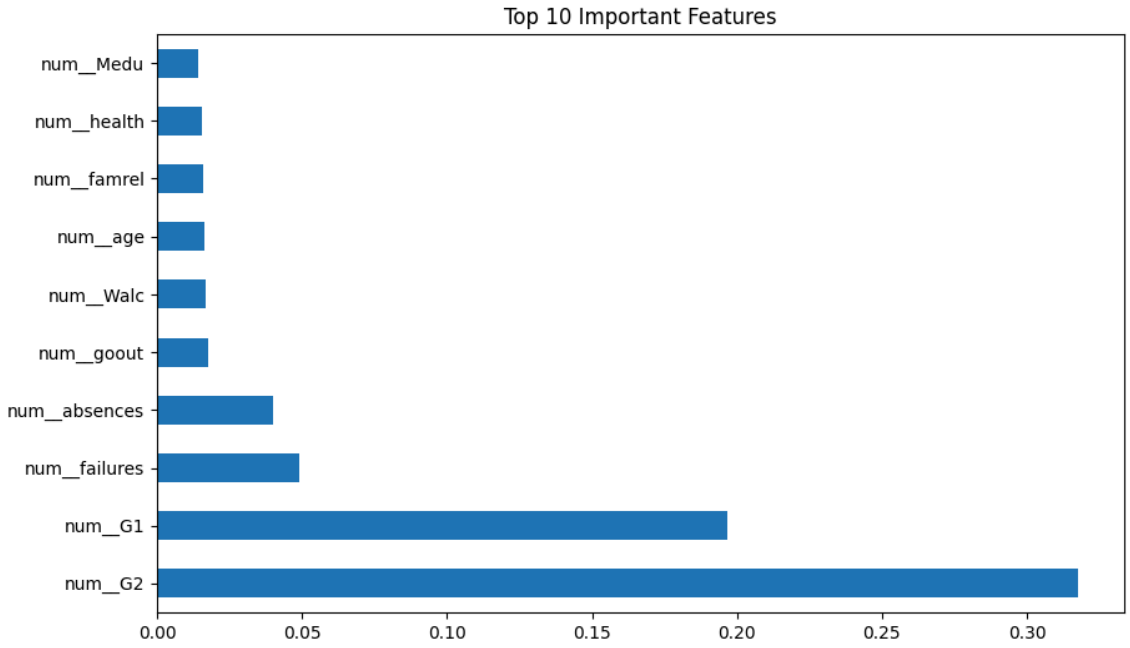

Figure 7: Feature Importance (Top 10 Features

The Random Forest model was employed to determine the most influential features in forecasting student performance. As depicted in Figure 7, previous grades (G2 and G1) emerged as the most significant predictors, followed by the number of past failures and student absences.

These findings underscore that historical academic performance is the most robust predictor of future success. Additionally, behavioural factors, such as attendance, also significantly influence outcomes.

5.7) Correlation Heatmap

To examine relationships between variables, a correlation heatmap was generated to visualize feature dependencies.

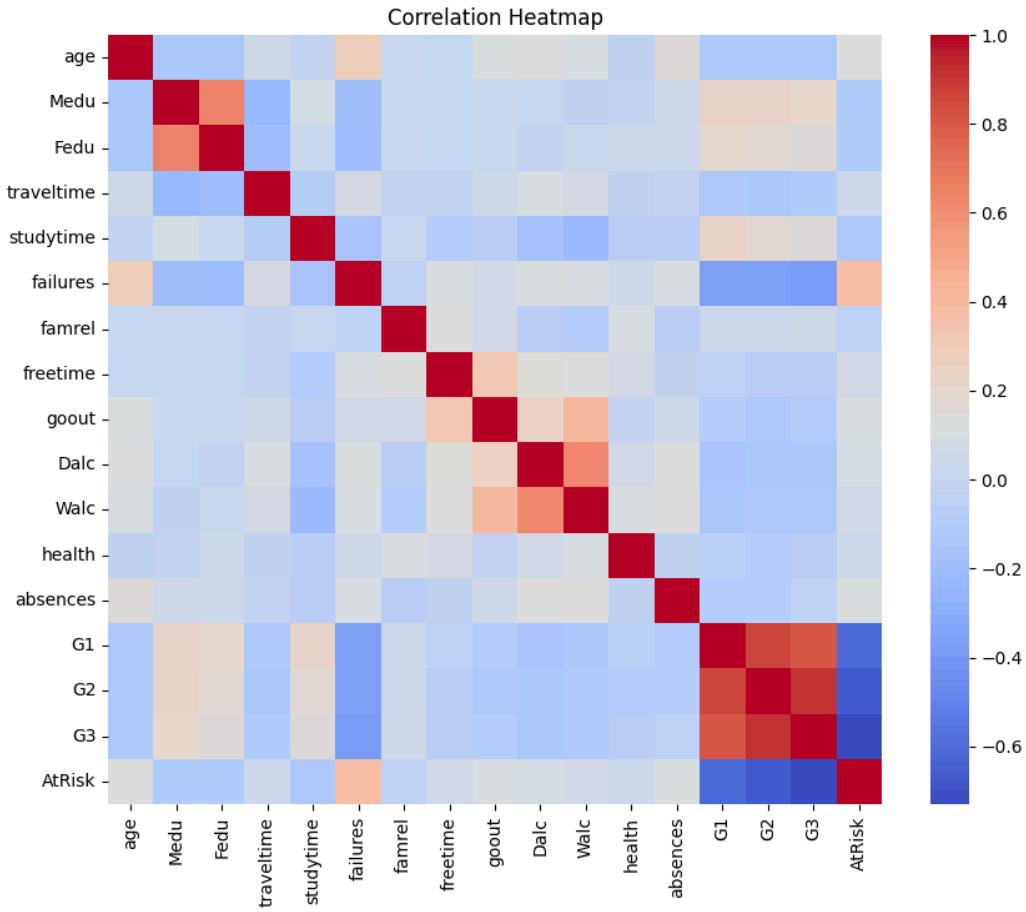

Figure 8: Correlation Heatmap

The heatmap illustrates the relationships between features. Strong positive correlations are observed between G1, G2, and G3, while the “At-Risk” label is negatively correlated with these grades. This supports the model’s reliance on prior academic performance.

6) Discussion

The findings indicate that both models effectively predicted student academic performance, each presenting unique benefits. Logistic Regression served as a reliable baseline due to its straightforward nature and clarity, achieving a marginally higher F1-score during cross-validation.

Conversely, Random Forest excelled in capturing intricate, non-linear relationships among features, suggesting that student performance is shaped by interactions among various factors rather than a strictly linear relationship.

For example, variables such as study time, absences, and prior grades interact in ways that ensemble models capture more effectively. As a result, Random Forest demonstrated robust performance on the test set and offered valuable insights through feature importance analysis.

The feature importance analysis indicated that prior academic performance (G1 and G2) had the most significant impact on predictions, followed by factors like failures and absences. This underscores that past performance is a strong predictor of future outcomes, while behavioural factors also hold substantial importance.

A significant challenge in this project was class imbalance, with fewer students identified as at-risk. To mitigate this, F1-score and stratified cross-validation were utilized, ensuring that model evaluation focused on accurately identifying at-risk students.

In conclusion, the results highlight a trade-off between interpretability and model complexity. While Logistic Regression provides simplicity and ease of understanding, Random Forest offers deeper insights into feature interactions and non-linear patterns.

7) Conclusion

This project has shown that machine learning models can successfully forecast student academic performance by utilizing structured educational data. Both Logistic Regression and Random Forest delivered impressive results, with Random Forest providing extra benefits in capturing non-linear relationships and pinpointing key predictive features.

The findings indicate that academic history, especially past grades, is crucial in predicting future performance. Additionally, factors like study time and attendance also influence outcomes.

These insights underscore the potential of data-driven approaches in education, allowing for the early identification of at-risk students and facilitating timely intervention strategies. Future research could investigate more advanced models or incorporate additional features to further enhance predictive performance.

8) Contribution Matrix

Name	**Student Number**	**Contribution**
Best Akinlabi	100837728	Wrote Introduction, Problem Statement, Dataset, Methodology, organized Results, report structuring
Ethan McLeod	100860608	Wrote code, got results and analyzed them from the code, got the datasets to use in the code
George Kassar	100872343	Wrote code, got results and analyzed them from the code, got the datasets to use in the code
Syed Nasir	100809447	Verified code implementations and goals alignments with proposal.